Actor Checkpoint Restoration in Asterix PPO: A Reference-Aware Study of Recovery Cost

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Abstract

A temporarily impaired reinforcement-learning agent can lag a clean control because of foregone progress, additional recovery cost, or both. We distinguish same-update gap attribution from interventions on detected harm relative to a historical pause benchmark. Our primary evidence is one preregistered Asterix PPO study with 32 new training seeds, reward-sign corruption and a fixed 200-update recovery horizon. Native pause-relative recovery cost is $-1.72\,[-1.94, -1.50]$ return points. Restoring only the pre-impairment actor, without the clean fork's additional progress, improves recovery by $+1.58\,[+1.37, +1.79]$. A single four-component practical-effect criterion, requiring native and late harm, rescue and reciprocal transfer, passes at $p = 3.32 \times 10^{-8}$, $\alpha = .01$. The intervention removes 91.8% of the measured mean cost (paired-bootstrap 95% interval 86.3–97.5%), not 91.8% of naturally stored damage. The interval jointly resamples paired training seeds conditional on a fixed evaluation batch. A small residual remains; matched-state practical equivalence fails. Specified optimizer, world-reset and second-evaluation checks retain positive rescue. A single donor/reference diagram and an integrated reproducibility appendix expose the counterfactual, uncertainty and artifact assumptions. Earlier studies retain their heterogeneous evidence status; there is no paper-wide multiplicity control. The conclusion is a setting- and intervention-specific repair result, not a general RL recovery law, proof of irreversible damage, a new-task plasticity result or a unique internal localization of impairment.

1 Introduction

A clean agent continues learning while an impaired copy falls behind. Replacing the impaired copy's parameters with the clean copy's later parameters may close this gap simply by importing the progress it missed. Such a transplant can identify a useful intervention without identifying a cause of recovery cost *beyond* lost training time. A reference trajectory is therefore part of the scientific claim, not a cosmetic plotting choice.

We ask a narrower question: *in one selected setting with detected pause-relative harm, does the learner's own pre-impairment actor checkpoint repair its subsequent recovery cost?* In Asterix PPO, a fixed 32-seed confirmation supports this intervention. The actor donor predates reward corruption; all other primary removal-time fields come from the naturally impaired learner. The primary requires four effects together, rather than selecting the most favourable endpoint. It is met, with native cost negative and actor rescue positive in all 32 seeds. The result is substantial but not complete restoration, and it does not locate a unique internal representation of impairment.

The main text presents the reference framework, this one study, its scope and the evidence hierarchy. The accompanying Supplementary Material preserves the full historical protocols, all result tables, negative/inconclusive primaries, context checks, registration timelines and deviations. Earlier experiments motivate the selected case; they are not pooled into its primary or treated as equally strong independent confirmations.

Contributions and scope. We provide an explicit reference-aware intervention framework; a fixed 32-seed confirmation that an older actor can repair measured recovery cost without extra clean-run progress; and

an auditable path from paired curves to statistical decisions and state-integrity checks. The contribution is neither the invention of parameter resetting nor a general algorithm guaranteed to preserve plasticity. Section 6 distinguishes our causal target from fresh resets, whole-checkpoint rollback and new-task plasticity studies. The present evidence remains one prior-selected Asterix PPO condition.

2 A reference-aware framework

2.1 Compact glossary

Table 1: Terminology used throughout. Each effect is conditional on its reference, future context, evaluation protocol and recovery horizon.

term	operational meaning
Tier I	Component-state intervention on the gap to a clean agent at the *same training update*. This gap includes foregone clean-run progress.
Tier II	A direct pause/onset-reference intervention coupled to detected *native pause-relative harm*. It supports an intervention-specific statement about that harm, not unique natural storage.
Pause reference	Here, the historical clean trace shifted back by the impairment duration: $H_i(k) = C_i(w_0+k)$. A matched onset-state learner is a separate intervention reference, not assumed identical to H.
Native recovery cost	The recovery AUC difference between the natural impaired continuation N_i and H_i. Negative means worse evaluation return. "Native" excludes an artificial clean-world continuation substituted for D.
State surgery	A specified field-level replacement of learner state, followed by ordinary learning. The causal effect includes downstream changes in actions, experience, critic and optimizer.
Shapley allocation	An allocation of a measured factorial gap: average each component's marginal effect over all insertion orders. It splits interactions by an ordering rule and sums to the selected gap; it is not an information decomposition.

2.2 One notation for donors, learner fields and contexts

We distinguish a saved runner state from the trajectory subsequently generated from it. For training seed i, P_i is the complete shared onset runner at update $w_0 = 390$; C_i^+ is the unimpaired fork at removal $w_1 = 440$; and D_i is the impaired runner at that same removal update, with corruption switched off. The superscript $+$ reminds the reader that C_i^+ contains 50 additional clean updates that are *not* available in P_i.

Write a composed removal state as $\mathcal{S}(a, v, o; x)$: a is actor parameters, v critic parameters, o the Adam state including its internal counter, and x every other numerical runner field. In particular, x_D retains the natural impaired environment, observations, future training random key, rollout-update counter, TrainState step and auxiliary fields. Keeping TrainState step is not the same as keeping Adam's internal count: the latter follows the selected o donor. Figure 1 gives the complete construction used by the primary contrasts.

$$N_i = \mathrm{Cont}\{\mathcal{S}(a_D, v_D, o_D; x_D)\}, \qquad R_i = \mathrm{Cont}\{\mathcal{S}(a_P, v_D, o_D; x_D)\}, \tag{1}$$

$$Q_{D,i} = \mathrm{Cont}\{\mathcal{S}(a_P, v_P, o_P; x_D)\}, \qquad B_{D,i} = \mathrm{Cont}\{\mathcal{S}(a_D, v_P, o_P; x_D)\}. \tag{2}$$

Here Cont means ordinary unrestricted recovery with the frozen PPO implementation, not an additional constraint that later fields remain equal. N is the native continuation; R the actor-restored continuation; Q_D the matched onset-*learner* reference; and B_D the reciprocal impaired-actor continuation. Q_D is neither the entire onset runner P nor the clean removal runner C^+.

The archive's three-letter codes select actor, critic and Adam donors in that order. In *pause cells*, C selects the onset donor P, not C^+. Thus N/R/Q_D/B_D correspond to DDD/CDD/CCC/DCC under natural D

context. The separate `fork/clean/donor/CCC` cell instead denotes the actual clean removal fork. These labels are an implementation convention, not interchangeable definitions of a clean reference. Appendix B lists all 11 cells and every preserved field.

2.3 A historical reference is not a unique literal pause

Let $C_i(u)$ denote the observed unimpaired evaluation trajectory at training update u. Our historical reference is the specified time shift $H_i(k) = C_i(w_0 + k)$, read at the same recovery index k as the impaired trajectory. At removal, $H_i(0)$ is the pre-impairment clean evaluation. At recovery $k = 200$, $H_i(200)$ uses clean update 590, whereas the native learner has reached update 640. This construction charges the impairment for the 50 updates it occupies rather than crediting the comparator with those extra updates.

There is a literal pause implementation that *can* realize this reference: freeze the *entire* runner P, including environment, random stream and all clocks, allow no external process to change it, and resume unrestricted training with the same deterministic, stationary program. Ignoring elapsed wall time, this produces the same clean continuation from P. This interpretation requires those specifications; "pause" alone does not determine them.

A paused learner resumed in the impaired learner's removal-time world is different. An environment might continue evolving during the pause, its data might be retained or discarded, and training/RNG/optimizer clocks might be frozen or advanced. Each choice defines another state and another counterfactual. Q_D is one explicitly executed learner-rollback intervention in x_D, not a claim that an actual pause would naturally produce x_D. Indeed, x_D was generated by the impaired learner's actions. No experiment equates these counterfactuals by naming both of them a pause.

Consequently, "native recovery cost" always means *native cost relative to the declared historical H benchmark*. It is not an intrinsic scalar attached to the impaired state independently of a reference. A small difference between the mean returns of Q_D and H can support a chosen practical margin in this assay; it does not prove identical states, identical trajectories or future-context neutrality. We therefore report repaired-minus-H and repaired-minus-Q_D residuals separately, including the failed latter margin.

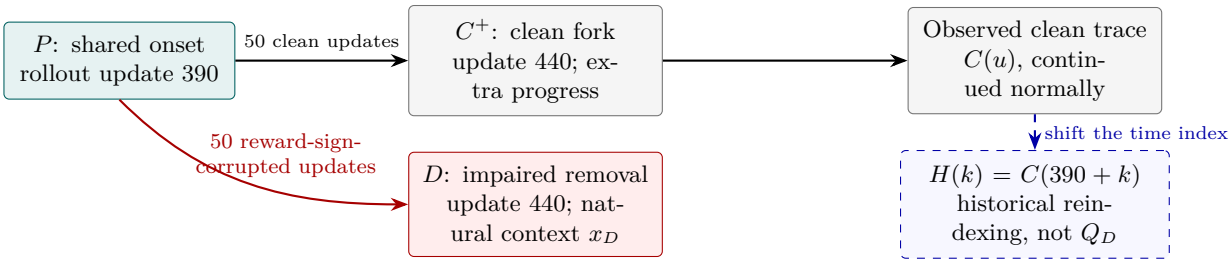
2.4 A total downstream intervention, not uniquely stored impairment

Let $\Phi^k(S)$ be the complete training continuation for k rollout updates from state S, including policy-dependent environment interaction and learner updates. The actor-restoration contrast compares evaluation of $\Phi^k\{\mathcal{S}(a_P, v_D, o_D; x_D)\}$ with evaluation of $\Phi^k\{\mathcal{S}(a_D, v_D, o_D; x_D)\}$. The root difference is actor parameters; the resulting effect is *total*, not restricted to the initial actor's direct contribution to a fixed dataset or fixed future state sequence.

The paths through changed action choices, visited states, rewards, critic estimates and optimizer updates are allowed to operate. After the first subsequent action, the experience used by the two learners need not coincide, even with common initial random keys. Our intervention does not block those mediators and does not estimate a controlled direct effect holding them fixed. It therefore establishes a useful causal handle under a specified background, not a natural decomposition of encoded information.

Nor does a successful actor transplant show that critic/optimizer state is uninvolved. Those fields may interact with the transplanted actor, and their conditional effects can change with the chosen donor or context. The smaller negative critic effect under common fresh Adam and the incomplete restoration residual are retained rather than absorbed into an "all damage is in the actor" conclusion. A larger reference-dependent repair ratio would still not identify a unique internal representation of impairment.

A Observed forks and a constructed reference



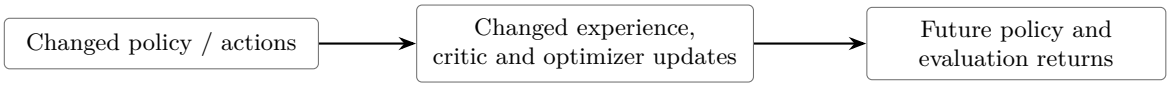
Repair uses P, not the extra-progress C^+. H reindexes the observed clean trace; it is not assumed to match a learner paused in the D context.

B Initial fields of the executed interventions

arm (archive code)	actor parameters	critic parameters	Adam state	other numerical runner fields
N (DDD)	a_D	v_D	o_D	Same x_D at removal: environment + observations, future PRNG key, rollout and TrainState clocks, all other auxiliaries. *Adam clock follows o.*
R (CDD)	a_P	v_D	o_D	
Q_D (CCC)	a_P	v_P	o_P	
B_D (DCC)	a_D	v_P	o_P	

In the pause-cell archive code, C means an onset donor from P, *not* the extra-progress clean fork C^+. Q_D rolls back learner parameters/moments but retains the natural impaired context; it is not the full runner P.

C Normal recovery after the splice: a total downstream effect



Native cost: $A(N) - A(H)$ Repair: $A(R) - A(N)$
Reciprocal transfer: $A(B_D) - A(Q_D)$

Solid arrows: observed/executed training. Dashed arrow: a reference readout. Matching holds *at removal*; future states and data are allowed to diverge. No arrow identifies uniquely stored impairment information.

Figure 1: **Donors, executed swaps and reference trajectories in one diagram.** A: P precedes impairment; the extra-progress C^+ actor never supplies the primary repair. The historical H trace is obtained by reindexing observations from the clean branch. B: all four primary continuations initially share x_D; colored fields show the P/D source of the actor, critic and Adam state. The code letter C denotes P only in these pause cells. C: actor replacement can change actions, experience and subsequent learner updates, so its effect includes these downstream paths. Solid arrows mark actual training/continuations; the dashed arrow marks a reference operation. Whole-runner freezing and learner rollback in the D world are different counterfactuals even when their mean returns are close.

2.5 What the reported AUC measures

Let w_0 be impairment onset, $w_1 = w_0 + L$ removal, and $V_i^X(k)$ the mean unrestricted evaluation return of seed i, arm X, at recovery checkpoint k. In the primary study,

$$\mathcal{K} = \{10, 20, \ldots, 200\}, \qquad A_i(X) = \frac{1}{20} \sum_{k \in \mathcal{K}} V_i^X(k). \tag{3}$$

Our "recovery AUC" is this **discrete time-average**, not an unnormalised integral or a trapezoidal area. It retains units of **evaluation episode-return points**. We average paired seed differences, not episodes or checkpoints as independent observations. The removal evaluation $k = 0$ is plotted but excluded from AUC. The late endpoint averages the last five checkpoints, $k = 160, 170, 180, 190, 200$.

The native cost and same-update gap use different benchmarks:

$$E_i^{\text{native}} = A_i(N) - A_i(H), \qquad H_i(k) = C_i(w_0 + k), \tag{4}$$

$$E_i^{\text{same}} = A_i(N) - A_i(C_{\text{same}}), \qquad C_{\text{same},i}(k) = C_i(w_1 + k). \tag{5}$$

A negative E^{native} is extra cost relative to this historical lost-time benchmark. It is not a guarantee that a literal paused runner in every future world would follow H. The matched onset-state reference is measured separately for precisely this reason.

Scale and practical meaning. An Asterix effect of -1.72 means that average evaluation return is 1.72 points below the specified reference across those 20 checkpoints. It is *not* 1.72 training steps, the total training reward lost, or a time-to-recovery estimate. One PPO update is 8,192 environment steps, so the 200-update recovery spans 1,638,400 steps. The registered .50 AUC margin was calibrated against a prior clean score of about 6.0 (roughly 8%); the .30 late margin was 5% of that same prior scale. These calibrations are specific to this game. Raw AUC magnitudes from SpaceInvaders, CartPole or PQN studies cannot be ranked or pooled as if they shared reward scale, evaluation policy and recovery grid. Cross-algorithm recovery comparisons additionally require matching *environment steps*, not update counts.

2.6 Intervention effects are not storage fractions

A clean-fork swap, a past-actor swap, a factorial main effect and a Shapley allocation answer different questions. The primary study uses a *background-specific* actor swap in the natural impaired context; it is not a full three-component allocation. Its repaired fraction is

$$\rho = \frac{\overline{A(R) - A(N)}}{-\overline{E^{\text{native}}}}, \tag{6}$$

where R denotes actor restoration. It is a ratio of mean intervention effects, not the average of seedwise percentages, a Shapley value, or the proportion of impairment encoded in actor weights. Changed future experience is part of the treatment effect, not held fixed by definition.

3 The preregistered Asterix study

Fixed selected condition. PPO (Schulman et al., 2017) uses separate 2×128 ReLU actor and critic MLPs, Adam at constant learning rate 5×10^{-4}, 64 environments × 128 steps, four epochs and eight minibatches. The environment is Asterix-MinAtar in gymnax (Young & Tian, 2019; Lange, 2022), implemented in JAX in the PureJaxRL style (Lu et al., 2022). Training rewards have their signs flipped during updates $[390, 440)$ (409,600 steps), followed by 200 unrestricted updates. Evaluation always uses true reward, clean observations and the full action set. Every ten updates, 64 stochastic-policy evaluation episodes use key 10000. The analysis pairs independent training seeds conditional on this fixed evaluation batch.

All 32 unused IDs 1200–1231 and all 11 requested cells were registered before target data. A previous independent eight-seed Asterix cohort motivated this selected condition; it is not pooled into the primary

Table 2: The single registered primary, $n = 32$. All four threshold-shifted tests and the consistency gate are required; $p_{\text{claim}} = 3.319 \times 10^{-8}$. Units are average evaluation-return points (Section 2.5). Brackets are nominal 95% paired-seed t-intervals.

contrast	mean [95% interval]	required mean	shifted one-sided p
Native AUC, $N - H$	$-1.722\,[-1.940, -1.504]$	$< -.50$	6.02×10^{-13}
Late native cost, $N - H$	$-1.187\,[-1.444, -.930]$	$< -.30$	3.32×10^{-8}
Actor-only rescue, $R - N$	$+1.580\,[+1.371, +1.790]$	$> +.50$	4.79×10^{-12}
Reciprocal transfer vs. Q_D	$-1.636\,[-1.828, -1.445]$	$< -.50$	1.39×10^{-13}

and does not make this a new-task generalization test. No low-competence seed was removed, no primary analysis occurred before all 32 completed, and no setting, endpoint, sample-size or evaluation-key switch was used.

Donors and natural continuation. P is the shared onset state at update 390, D the impaired state at 440, and C^+ the unimpaired fork at 440. The main repair replaces *only actor parameters in D with those from P*; it does not use the extra-progress actor from C. The removal-time critic, Adam state, environment, observations, future RNG and counters remain those of D. The all-impaired cell is asserted equal over the complete runner-state tree to untouched natural D. Actor/critic/optimizer composition is explicit in code; after intervention these fields and collected experience can diverge normally.

For reciprocal transfer, the impaired actor is inserted into a learner with onset critic parameters and Adam state, under the same natural D world/RNG/counter context. We denote that matched learner reference by Q_D: it is *not* the full onset runner P or the historical trajectory H. In archived pause-cell codes, C means an onset donor and D an impaired donor: DDD is native, CDD actor-restored, CCC the matched reference, and DCC reciprocal impaired-actor transfer.

One primary claim, four required parts. The registration fixes four one-sided paired-t tests against practical-effect thresholds (Table 2). The single intersection–union claim uses $p_{\text{claim}} = \max_j p_j$ at $\alpha = .01$, with an additional requirement of at least 24/32 native-negative and 24/32 rescue-positive seeds. All parts must pass. Intervals are nominal training-seed intervals, with 99% intervals, exact signs, paired bootstrap and simultaneous-bound sensitivities retained in the supplement and analysis artifact (Henderson et al., 2018; Agarwal et al., 2021). The fixed second evaluation key 54321, common-fresh-Adam factorial and reset-world rescue are prespecified *secondary* checks; they cannot replace a failed primary.

4 Results: most cost repaired, not full restoration

The compound criterion is met. Native AUC is negative in 32/32 seeds and actor rescue positive in 32/32, exceeding the registered 24/32 requirements. The late mean clears its practical margin; 31/32 late contrasts are negative, not every seed. Figure 2 shows every primary seed effect, both reference trajectories and the fixed robustness checks.

The past-actor intervention removes **91.8% of the mean native historical-pause cost**, with paired-bootstrap 95% interval 86.3–97.5%. This number describes the ratio ρ in this one experiment. It is neither "91.8% of damage stored in the actor" nor a universally applicable restoration rate.

Robustness does not imply neutrality or full restoration. All secondary results remain visible in Table 3. Fresh Adam, a reset world and the second evaluation key retain positive rescue intervals. The smaller negative fresh-Adam critic effect is not declared absent. Neither a nonsignificant context difference nor positive rescue in another context establishes context neutrality. The second key is not selected or pooled into the primary.

The repaired-minus-historical residual is small but negative. Its secondary $\pm.30$ TOST margin passes, while the same margin against Q_D *fails* (Lakens, 2017). Practical-margin agreement is not literal equality.

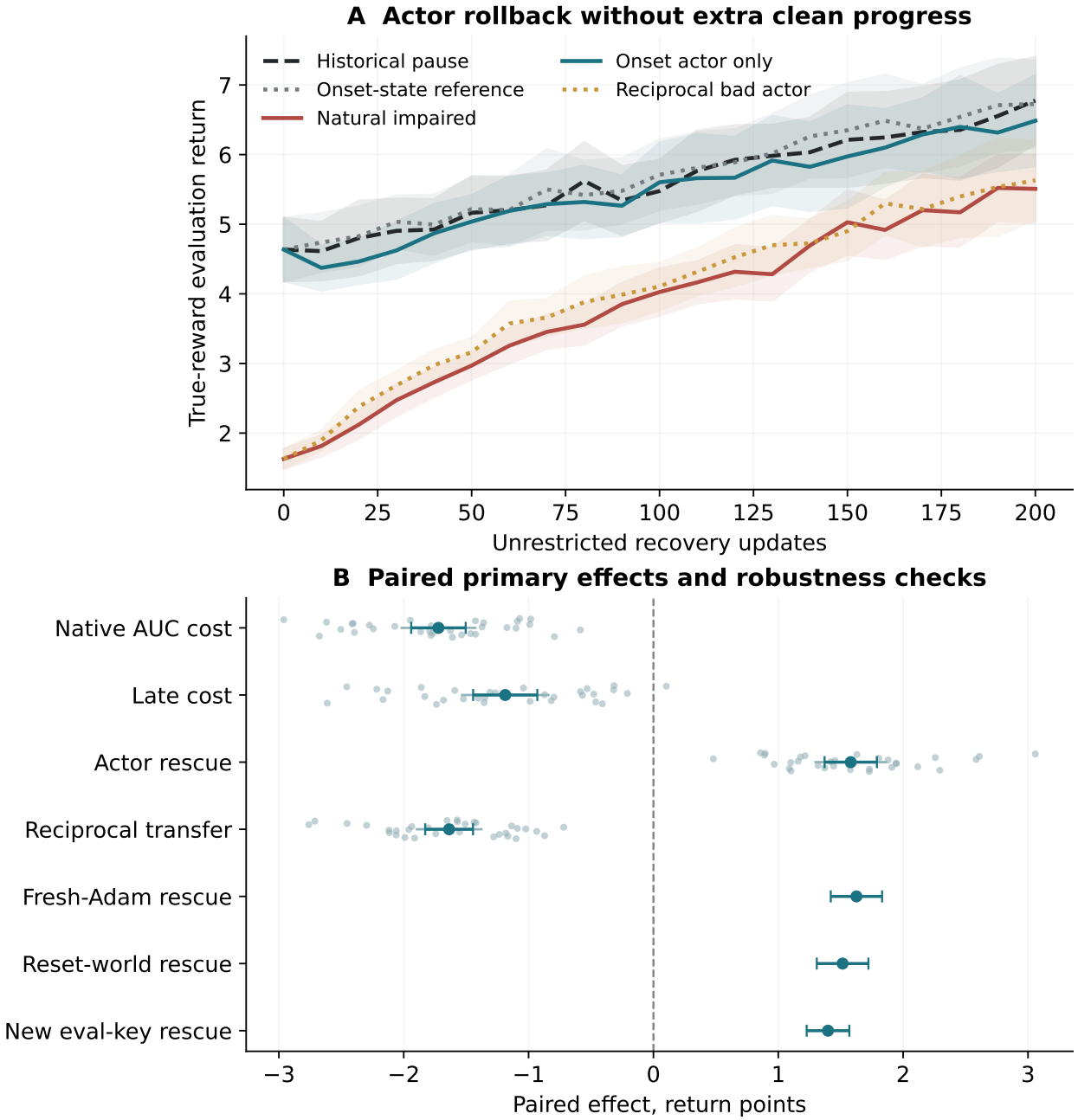


Figure 2: **One selected-condition actor-restoration result.** A: evaluations use true reward; the historical pause trace, matched onset-state reference, native learner, onset-actor repair and reciprocal actor transfer are distinct arms. The plotted removal point is excluded from AUC. Bands are nominal pointwise 99% seed intervals, not simultaneous trajectory bands. B: all 32 paired primary effects (small points), means and 95% intervals, with thin 99% primary whiskers. Secondary rescues are shown without promoting them to additional primary claims. Worlds and non-actor fields match *at removal* for the primary repair; downstream trajectories may diverge.

Table 3: Prespecified secondary checks, $n = 32$, nominal 95% paired-seed intervals. TOST uses the same $\pm.30$ practical margin for both repair residuals; its outcome depends on the reference. These checks are not a new multiplicity-controlled primary family.

contrast	mean [95% interval]	interpretation
Fresh-Adam actor rescue	$+1.626\,[+1.420, +1.832]$	positive rescue interval
Fresh-Adam critic main effect	$-.129\,[-.229, -.029]$	smaller, not absent
Reset-world actor rescue	$+1.515\,[+1.308, +1.722]$	specified alternative context
Second-key actor rescue	$+1.398\,[+1.227, +1.569]$	key 54321; not pooled
Restored minus historical H	$-.141\,[-.246, -.037]$	TOST $p = .00211$: margin met
Restored minus matched Q_D	$-.231\,[-.370, -.091]$	TOST $p = .160$: not met
Matched Q_D minus H	$+.089\,[-.035, +.213]$	practical agreement, not identity

In particular, positive rescue and a 91.8% reduction do not establish complete recovery, even though the primary compound passes.

5 What the statistical uncertainty does and does not cover

5.1 The inferential unit is a training seed, conditional on evaluation randomness

Separate three random objects: the initialization/training stream indexed by seed S, the evaluation random tape U, and the explicitly selected scientific condition (game, algorithm, architecture, corruption and horizon). For a fixed evaluation tape U_0, each trained policy produces one 64-episode mean at each checkpoint. The paired primary row is

$$Z_i(U_0) = \big(E_i^{\text{native}},\ E_i^{\text{late}},\ A_i(R) - A_i(N),\ A_i(B_D) - A_i(Q_D)\big). \tag{7}$$

The reported primary intervals estimate across-training-seed means of these measured quantities *conditional on U_0 and the selected condition*. They do not integrate over a redrawn evaluation population. Independent training streams change initialization, sampled training experience and learned parameters together; this design does not separately estimate their individual variance components.

The evaluation function resets 64 environments and follows the first episode in each, capped by the environment's episode limit. Its fixed key determines reset, stochastic-policy and transition random draws. The same key 10000 is reused across arms, checkpoints and training seeds as common random numbers. Pairing can reduce variance of contrasts, but it does not make a finite evaluation batch equal to the population performance of a policy. We do not treat 32 seeds $\times$ 64 episodes $\times$ 20 checkpoints as independent observations.

For a seed contrast Z, the nominal two-sided interval is

$$\bar{Z} \ \pm\ t_{31,1-\alpha_c/2}\, s_Z/\sqrt{32}, \tag{8}$$

with $\alpha_c = .05$ or .01 for the displayed 95% or 99% interval. These levels differ from the primary compound's $\alpha = .01$ decision rule. The usual seed-level sampling assumptions remain; the intervals are not distribution-free guarantees. Exact signs, mean-bootstrap sensitivity and simultaneous one-sided bounds qualify the result but do not turn it into a paper-wide test. Trajectory bands in Figure 2 are *pointwise*99% intervals, not a simultaneous confidence band over time.

Table 4: Scope of the reported intervals. Reusing fixed evaluation randomness is a design choice, not a claim that evaluation uncertainty is zero.

source or dependency	represented in the reported seed intervals?
Initialization and training-stream differences	Yes, jointly through 32 independent training-seed rows, at fixed evaluation tape.
Shared history and paired arms	Yes; differences are formed within seed before inference.
Dependence among recovery checkpoints	Kept inside a seed's AUC/late summary; checkpoints are not resampled as independent units.
A newly drawn 64-episode evaluation tape	No; primary conditioning fixes U_0.
Selection of task, architecture, hyperparameters or horizon	No; one selected condition is held fixed.
A literal pause in a different future context	No; that changes the estimand, not merely its sampling error.

5.2 What the second evaluation key establishes

Key 54321 was frozen before the new study and evaluated on the specified clean, native and actor-restored cells. It gives native AUC $-1.599\,[-1.762, -1.436]$, late cost $-1.070\,[-1.315, -.825]$ and actor rescue $+1.398\,[+1.227, +1.569]$. These are separate conditional-on-U_1 estimates for the *same* 32 trained seeds. The second key is not selected after comparing outcomes, pooled with U_0, counted as another training cohort, or used to replace a failed primary.

Agreement across these two specified tapes is useful robustness evidence. It does not provide an interval over all possible evaluation worlds/policy draws, and it does not establish generalization across tasks. A future evaluation-marginal analysis would require an explicit design with enough independently generated evaluation tapes and an uncertainty calculation respecting the crossed/common-random-number structure. We do not retrospectively claim to have performed that analysis.

5.3 The repair ratio requires a joint, paired bootstrap

Let $B_i = -E_i^{\text{native}}$ be the measured positive historical-pause cost and $Y_i = A_i(R) - A_i(N)$ the rescue. The reported ratio is $\hat{\rho} = \bar{Y}/\bar{B}$, not $\overline{Y_i/B_i}$. Numerator and denominator share the native trajectory, the initial seed and the evaluation tape. Their uncertainty must not be combined as if they were independent estimates.

For each of 20,000 bootstrap draws, we sample 32 seed indices with replacement *once* and use that same index vector for both Y and B:

$$\rho^{*(b)} = \frac{32^{-1}\sum_{j=1}^{32} Y_{I_j^{(b)}}}{32^{-1}\sum_{j=1}^{32} B_{I_j^{(b)}}}. \tag{9}$$

Resampling complete paired rows preserves their empirical covariance and the already-aggregated within-seed checkpoint/arm dependence. The frozen implementation uses NumPy's default generator seed 140033 and the linear 2.5th/97.5th percentiles. It neither clips ratios to $[0, 1]$ nor substitutes a fixed denominator. The resulting interval is 86.3–97.5%, conditional on the same evaluation tape and condition as the inputs. Appendix C gives executable pseudocode and an exact reproduction check.

The covariance matters. A delta-method expansion, provided only to explain the joint calculation, is

$$\operatorname{Var}(\hat{\rho}) \;\approx\; \frac{\operatorname{Var}(Y) + \rho^2 \operatorname{Var}(B) - 2\rho \operatorname{Cov}(Y, B)}{n\,\mu_B^2}. \tag{10}$$

Independently resampling numerator and denominator would discard the last term. We do not report that as the paper's interval. Every observed B_i is positive here, so empirical bootstrap denominators are positive; this is not a general guarantee for ratio estimands in other settings. Ratios may in principle exceed 100% under over-repair or become unstable near a zero mean cost. The interpretation remains reference-dependent: changing H changes B and therefore ρ, even when the direct actor-rescue contrast is held fixed.

Finally, the 91.8% ratio and its bootstrap interval are a descriptive repair summary, not another required component of the four-part primary test and not a proportion of impairment information stored in a component. The negative residual and failed matched-state restoration margin remain part of the conclusion.

6 Why this result is different from a generic reset or plasticity claim

The empirical advance is deliberately narrower than a new universal recovery principle. In one selected setting, the complete registered conjunction demonstrates native historical-pause cost, a late-window cost, repair by an *older* actor, and reciprocal impaired-actor transfer. The older donor is important: the repair cannot be explained by copying the extra clean progress accumulated during impairment. Retaining the actual impaired critic, Adam and removal-time world also makes this a different intervention from resetting or rewinding the whole learner.

That distinction is scientifically useful even without task-wide generality. It separates two explanations that an ordinary clean-fork transplant cannot distinguish: importing foregone progress versus changing a state variable that alters recovery relative to a specified lost-time benchmark. It does *not* distinguish a direct actor-information effect from effects mediated by future actions/data, and it does not turn the selected benchmark into the unique correct counterfactual. The reference-aware assay is the methodological contribution; the 32-seed result is an independent confirmation *within a prior-selected condition*, not a prevalence estimate.

6.1 Fresh resets, retained competence and plasticity are different targets

The primacy-bias intervention periodically reinitializes network components while retaining replay experience (Nikishin et al., 2022). That tests the usefulness of discarding a fitted state while relearning from retained data. Our primary PPO assay has no replay buffer and restores a known past actor rather than a fresh random actor. These are different treatments: past competence is available in one and may be discarded in the other. The old four-seed SpaceInvaders reset result does not establish that past-actor restoration outperforms fresh resets in this Asterix study.

Plasticity injection changes trainable state while preserving predictions at intervention time (Nikishin et al., 2023); probe-task work directly examines ability to fit new targets (Lyle et al., 2023). On-policy plasticity studies examine domain shifts and find that remedies can behave differently from the off-policy setting (Juliani & Ash, 2024). Our actor restoration *does* change predictions and future behaviour, and the learning objective returns to the same task. Good same-task recovery therefore cannot substitute for a measurement of capacity on a new task.

Likewise, module-specific conclusions need not transfer. In their visual-RL settings, Ma et al. (2024) identify critic plasticity as a central bottleneck. Our result identifies a useful actor intervention in a different algorithm/impairment/context. Neither finding licenses a universal ranking of actors and critics. Transient non-stationarity and representation-repair work (Igl et al., 2021), and broader studies of plasticity loss (Dohare et al., 2024; Klein et al., 2024), motivate these questions without making the present recovery contrast a representation diagnostic.

Table 5: Interventions should be compared by what they change and what their outcomes establish. This is a conceptual distinction, not a head-to-head performance benchmark.

operation	what is supplied or discarded?	what is not established merely by improvement?
Later clean-actor transplant	An actor with additional clean progress during impairment.	Harm beyond that foregone progress, or an intrinsic damage carrier.
Past-actor restoration (here)	The learner's pre-impairment actor; current non-actor fields retained.	Unique storage, new-task plasticity, complete restoration, or superiority to other recovery methods.
Fresh partial/full reset	New initialized parameters; replay or other knowledge carriers may remain.	That the discarded state lacked useful competence, or that the same reset helps another algorithm/context.
Plasticity injection / new-target probe	Altered trainability or an explicit test of fitting new targets.	That the present same-task pause-relative cost has the same mechanism.
Whole-runner rollback	Earlier parameters, optimizer, world, random stream and clocks together.	The actor-specific intervention effect under the *current* natural impaired context.

6.2 The significance and the boundary of the contribution

The result closes a specific interpretive gap: an onset checkpoint with no extra-progress advantage can repair most measured native recovery cost under the declared intervention. Reciprocal transfer and the fixed fresh-Adam, reset-world and independent-key checks constrain simple alternative explanations. The 32-seed design, full paired data and state-integrity audit make this a stronger claim than an exploratory four-seed allocation.

There is no matched Asterix comparison against plasticity injection, distillation, a fresh-reset strategy or an automatically selected checkpoint. We therefore make no efficiency, safety or state-of-the-art claim over those methods. Selecting the right checkpoint, detecting an impairment online, trading preserved competence against future learnability, and assessing larger/shared-encoder agents remain open problems. The reward mechanism, separate-tower MLP architecture and recovery horizon are each held fixed, not demonstrated irrelevant.

7 Evidence hierarchy across the research record

The primary result above is the strongest evidence for the *named intervention*. The broader record is heterogeneous in sample size, design status, reference and inferential target. Table 6 prevents a favourable result in one tier from upgrading an unresolved result in another. A preregistered design is not automatically a successful outcome. New contrasts on reused histories are not independent-seed replications, and selected eligible conditions are not a task-prevalence sample.

Table 6: Evidence hierarchy. Status attaches to the listed estimand and study, not to a universal carrier mechanism. Full values, references, N, families and deviations remain in the supplement and machine-readable inference map.

status	evidence and boundary
Strongest registered support	The present 32-seed Asterix PPO compound meets its fixed $\alpha = .01$ rule: native and late pause-relative harm, past-actor rescue and reciprocal transfer. One setting and one horizon.
Context-specific support	Prior independent Asterix 8 (Tier II) and actuator-noise 8 (Tier I) meet their fixed five-claim Phase-13 Holm criteria. The latter does *not* show native pause-relative harm. Both are selected conditions, not a task-prevalence sample.
Exploratory / descriptive	Early timing and four-seed reset/drift studies are exploratory. PPO's pooled gap allocation ($n = 12$) includes four pilot and eight new registered IDs. PQN-R network-block tests ($n = 12/8$, parameters plus RAdam) remain exploratory; Freeway has only 4 complete of 8 planned. Reused histories are not fresh-seed replications.
Narrow practical bound	Matched replay identity, $n = 32$: selected ± 1 TOST $p = .0495$; $\pm .75$ fails. Smaller effects remain possible. This is not absence of a data effect.
Unresolved primaries / mechanisms	Independent CartPole 8's stronger compound fails Phase-13 correction ($p = .0931$). The controlled-label assay 8 meets neither its substantial-harm nor equivalence rule. A buffer-through-learning mechanism remains interpretive; no secondary contrast rescues these primaries.
Prospective utility not validated	Held-out 16 frozen acute+KL prediction has MSE 68.78 versus 32.27 for acute-only and fails its utility criterion. Earlier KL associations do not establish useful future-failure prediction or a causal mechanism.

No paper-wide error-control claim. The 32-seed compound is one separate registered $\alpha = .01$ claim. Only the five Phase-13 primaries share their specified Holm family (Holm, 1979); older studies have other local families, exploratory analyses and disclosed interim looks. We neither combine these into a retrospective paper-wide false-positive guarantee nor count favourable panels as exchangeable independent confirmations. The main SpaceInvaders movement result, for example, has historical-pause AUC $+1.9\,[-.7, +4.4]$, $n = 12$: harm worse than -3 is excluded, but equivalence is not established. Its 90% actor allocation concerns a Tier-I gap, not demonstrated beyond-time harm.

8 Direct reproduction and artifact availability

The primary reproduction guide links every group, field, command and availability limit. Table 7 gives direct entry points. The protocol, runner, PPO and composer match registration 6262b2b byte for byte. Pinned dependencies and manifests record the numerical environment and source revisions; analysis source is linked at frozen release 74c6ceb.

Reanalysis is not rollout replay. All original primary seed-level means, reference alignment data, statistical source and recorded integrity checks are available. The original experiment did not retain full serialized donor trees. Appendix E distinguishes that fact from later, checked numerical-state reconstructions and gives their actual completion/publication status. A fingerprint is not a checkpoint download. Loading a validated numerical donor can avoid retraining its prehistory, but regenerating its recovery trajectory or evaluating an-

Table 7: Direct primary-study artifacts. Frozen scientific-source links point to release 74c6ceb; new CSV/index links point to the presentation release. No donor checkpoint blob is implied by a fingerprint.

item	directly linked artifact
Protocol and statistical rule	Registration, registered analysis implementation
Donor construction and swaps	Onset/fork construction and natural-context composition; actor/critic/Adam field selection
State-integrity checks	Runtime full-tree identity assertions; independent arithmetic/source audit
Every paired training seed	32-row primary CSV; full primary/secondary values and sensitivities
Eight raw groups and manifests	Hashed artifact index, including recorded checks and P/C/D *batch* fingerprints

other random tape still requires computation. The self-contained appendix supplies the cell map, state fields, joint bootstrap, complete paired seed table and restoration commands; the separate historical supplement is not needed for primary statistical reanalysis.

9 Limitations and conclusion

What the result establishes. Under one declared natural future context, a learner's own pre-impairment actor checkpoint causes a substantial improvement in measured pause-relative recovery. Reciprocal transfer and the specified checks constrain a simple extra-clean-progress or optimizer/world-only explanation. The repair removes most, not all, of this mean cost. It remains a total downstream effect of this field-level intervention; it is not a causal accounting of uniquely stored impairment information.

What it does not establish.

- **Generality:** the strongest confirmation is one Asterix environment implementation, PPO, deterministic reward-sign corruption, small MLPs and one 200-update recovery horizon. It is not a law for RL agents, other architectures, impairments or training budgets.
- **Context independence:** historical pause and matched learner references are different; natural, standardized and reset futures answer different interventions. Positive checks do not establish a context-neutral decomposition.
- **Irreversibility or new-task plasticity:** an AUC cost and a late-window deficit do not prove permanent damage. No experiment measures preserved capacity to learn a new task after impairment.
- **Mechanism or storage:** actor replacement changes future data and learning. Replay time courses, inconclusive label-exposure primaries and policy-KL associations do not identify a unique internal representation or natural storage location of impairment.
- **Collective certainty or deployment:** heterogeneous local test families provide no paper-wide error guarantee or task-prevalence estimate. The fixed evaluation pools and unavailable full donor trees limit reproduction; no automatic checkpoint-selection or deployment rollback policy is validated.

The defensible conclusion is specific: *a past-actor intervention repairs 91.8% of the measured mean native recovery cost in this registered Asterix PPO experiment*, with a small residual and failed matched-state practical equivalence explicitly retained. The interval is conditional on the fixed evaluation batch and jointly resamples numerator/denominator seed rows; changing the reference changes the repair fraction. The findings do **not** establish irreversible damage, preserved plasticity on new tasks, or a unique internal representation of impairment. The framework is useful because it makes these boundaries explicit, not because it turns one successful restoration into a general recovery law.

Broader impact statement

These are controlled small-agent experiments in simulated environments. Potential use in reliable RL recovery requires evaluation of checkpoint selection, new tasks, longer horizons and deployment risks; no safety guarantee follows from this study.

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A Reproducibility map and frozen scientific specification

This appendix is part of the main PDF. It contains the primary protocol, complete cell map, numerical estimators, uncertainty algorithm, all paired primary seed results, state-integrity rules, artifact tiers and commands. The separate historical supplement is useful for the earlier research record but is *not required to reconstruct the primary statistical result.*

A.1 Which release is frozen?

The public scientific release 74c6ceb contains the completed original 32-seed results. The later presentation release a21654c changes their exposition, not the primary numerical artifact. Registration 6262b2b fixes the primary scientific protocol and core runner/PPO/composition files. The analysis implementation was added before formal target inference; it is linked at the completed scientific release, not falsely claimed to be a file already present in the initial registration commit.

The reconstruction wrapper is a separate instrumented program. It imports the unchanged scientific runner and checks its four registered source files byte for byte. Serialization and evaluation recording do not change the registered reward corruption, training budget, cells or inferential rules. Reconstructed arrays are labelled with their later creation date and original numerical fingerprints. They are not retroactively described as files retained when the original experiment ran.

Table 8: Complete primary configuration. These choices are held fixed; their appearance in a table does not establish invariance to other choices.

item	value
Environment / implementation	Asterix-MinAtar, gymnax 0.0.9; flattened observations
Agent / architecture	PPO; separate actor and critic MLPs, two 128-unit ReLU hidden layers each; no shared encoder or recurrence
Initialization	Orthogonal hidden kernels (gain $\sqrt{2}$); actor output gain .01; critic output gain 1; zero biases
Optimizer	Adam, learning rate 5×10^{-4} constant, $\epsilon = 10^{-5}$; global gradient clip .5
Rollout and optimization	64 environments $\times$ 128 steps; 8,192 steps/update; 4 epochs $\times$ 8 minibatches (1,024 samples/minibatch)
Discount / advantage trace	$\gamma = .99$, GAE $\lambda = .95$
PPO losses	Clip .2; entropy coefficient .01; value coefficient .5
Training corruption	Reward sign reversal with probability 1 during $[390, 440)$; observations/actions otherwise unchanged
Training before/after removal	Shared 390-update prehistory; 50-update impaired/clean fork; 200 unrestricted recovery updates
Evaluation	Every 10 rollout updates; 64 first-episode returns, full actions, true reward, stochastic policy, environment episode cap
Primary / secondary tapes	Fixed key 10000; fixed secondary 54321 in the three specified cells
Independent training IDs	1200–1231, in the eight fixed four-seed groups; none filtered
Primary family	One four-component intersection–union, $\alpha = .01$, plus the two 24/32 consistency requirements
Secondary restoration margin	$\pm.30$ against both historical H and matched Q_D; not literal zero
Hardware / original full runtime	Two CPU cores, about 2GB RAM, no GPU; 6h34m33s for the original eight groups

Each training seed controls initialization and its training randomness. The original runner records package versions, configuration, seed group, source revision, status and whether the tree was clean. The fixed corruption is evaluated without corruption: negative training rewards must not leak into evaluation. "Late" refers to the previously selected competent onset, not to an onset sweep or a universal critical period in this 32-seed study.

A.2 Packages and numerical environment

The full version list is in `requirements-repro.txt`. Core versions are Python 3.13, JAX/jaxlib 0.11.2, Flax 0.12.10, Optax 0.2.8, gymnax 0.0.9, NumPy 2.5.4 and SciPy 1.18.1. Architecture and initialization are read from the frozen source, not inferred from plotted returns. Each reconstructed checkpoint manifest additionally lists all numerical leaf shapes/dtypes, so observation/action-dependent layer dimensions can be checked rather than guessed.

Matching package versions is necessary but not a promise of identical numerical results on every backend or compiler/CPU configuration. The reproduced state fingerprints and trajectory checks are the actual validation criterion. If they fail elsewhere, preserve and report the discrepancy rather than adjust scientific source or select favourable seeds. Run JAX jobs sequentially on a small-memory CPU machine; concurrent JAX compilation/tests can exceed the memory budget.

B Every intervention cell and preserved state field

B.1 The complete 11-cell map

For the primary, N, R, Q_D, B_D always start from the natural D context. "Donor" below means selecting the corresponding donor Adam block, not copying all clocks from that donor. A fresh-Adam cell overwrites that block with the common fresh optimizer state after actor/critic composition.

Table 9: All registered cells. P means onset; D means impaired removal; C^+ means clean removal;0 means common fresh Adam. The external context includes all non-train-state fields and the retained TrainState step.

archive cell	actor	critic	Adam	context	role / evaluation tapes
`fork/clean/donor/CCC`	C^+	C^+	C^+	x_{C^+}	clean trace;10000/54321
`pause/damaged/donor/CCC`	P	P	P	x_D	Q_D;10000
`pause/damaged/donor/DDD`	D	D	D	x_D	native N;10000/54321
`pause/damaged/donor/CDD`	P	D	D	x_D	repair R;10000/54321
`pause/damaged/donor/DCC`	D	P	P	x_D	reciprocal B_D;10000
`pause/damaged/zero/CCC`	P	P	0	x_D	fresh-Adam factorial
`pause/damaged/zero/CDC`	P	D	0	x_D	fresh-Adam factorial
`pause/damaged/zero/DCC`	D	P	0	x_D	fresh-Adam factorial
`pause/damaged/zero/DDC`	D	D	0	x_D	fresh-Adam factorial
`pause/reset/donor/DDD`	D	D	D	$x_{D,r}$	reset-world native;10000
`pause/reset/donor/CDD`	P	D	D	$x_{D,r}$	reset-world repair;10000

The literal letter C has different implementation meanings in the fork and pause reference modes, which is why the main text uses $\mathrm{P}/C^+/N/R/Q_D/B_D$ instead of inferring donor age from the letter alone. The reset context $x_{D,r}$ replaces only environment state and observations, using initialization seed $900000{+}s$. Its future training key, counters, reward bank, masks and all other fields remain D. Common fresh Adam is initialized from the same optimizer transform in all four actor–critic combinations; it does not define a natural decomposition of original optimizer damage.

B.2 Field-level integrity, including clocks

The actor keys are `a0`, `a1`, `Dense_0`; critic keys are `c0`, `c1`, `Dense_1`. They form separate towers. The generic composer selects these parameter groups and the Adam state, but its returned runner context is *not* by itself the final intervention. The primary runner takes the composed TrainState and wraps it in the explicitly chosen D context. Both code locations are linked in the main artifact table.

There are 32 gradient updates per rollout update (four epochs times eight minibatches). Rollout counters and gradient-step/Adam counters must therefore not be conflated. The source retains the D TrainState step

Table 10: What is preserved for the actor-only repair R at removal. Field identities hold at the splice, not throughout subsequent training.

field group	repair R / matched reference Q_D
Actor parameters	P / P
Critic parameters	D / P
Adam moments and internal optimizer count	D / P
TrainState step	D / D; distinct from the Adam internal counter
Environment state and observations	D / D
Future training PRNG key	D / D initially; policy-dependent experience can diverge
Rollout-update counter	D / D, at update 440
Deficit bounds	Disabled after removal in every recovery cell
Reward bank, action mask	D / D
Behaviour-policy auxiliary actor, cutoff, temperature and ratio flag	D / D; the optional behaviour swap is inactive in this assay
Anchor actor and other numerical auxiliaries	D / D; no drift/freezing intervention is active

even when onset Adam is selected for Q_D or B_D. A literal whole-runner P freeze retains all P clocks instead. This is another reason those states are not identical.

The original summaries record full-tree equality assertions for native DDD versus untouched D and clean fork CCC versus untouched C^+. They do not contain all intermediate CDD runner objects for retrospective inspection. The reconstruction additionally checks every composed actor/critic/Adam donor, all non-train-state D fields and the retained TrainState step. Those later checks are labelled as recomputed checks, not falsely attributed to an originally saved field audit.

C Exact estimators, tests and the joint ratio bootstrap

C.1 Array orientation and historical alignment

Each `cells[key].eval_return_mean` matrix is seed-by-checkpoint: four rows and 21 columns per original group. Column 0 is removal, then +10 through+200. Fork arrays in `fork_eval.C/D` and `fork_eval_alt.C` instead have checkpoint-by-seed orientation. The CSV export gives one row per training seed, not per episode or checkpoint.

For an absolute recovery evaluation update $u = 440 + k$, read the historical clean mean at $u - 50 = 390 + k$. The clean bank is formed by the fork evaluations 390–440 and the continuing `fork/clean/donor/CCC` cell 440–640. Where the two meet at 440, their clean evaluations agree. A shift by 40 or using the same update would define a different comparison. The independent verifier reconstructs this alignment from raw rows rather than trusting a precomputed native-cost array.

For a curve X with 21 checkpoint values, $A(X) = \mathrm{mean}(X[1:])$. The late comparison is the mean of the five *paired differences* at 160–200. All four primary vectors are formed within seed before averaging:

$$\gamma_i = A_i(N) - A_i(H), \qquad \ell_i = \tfrac{1}{5} \sum_{k=160,170,\ldots,200} \{N_i(k) - H_i(k)\}, \tag{11}$$

$$\tau_i = A_i(R) - A_i(N), \qquad \delta_i = A_i(B_D) - A_i(Q_D). \tag{12}$$

Episode-return units are preserved. An unnormalised sum or an area including removal would change the scale and is not what the registered .50/.30 thresholds test.

C.2 Shifted practical-effect tests and consistency

The one-sided alternatives are $\mathbb{E}\gamma < -.50$, $\mathbb{E}\ell < -.30$, $\mathbb{E}\tau > +.50$ and $\mathbb{E}\delta < -.50$. Implement them by paired-seed one-sample t tests of $\gamma + .50$, $\ell + .30$, $\tau - .50$ and $\delta + .50$ against zero, with the corresponding less/greater direction. Use 31 degrees of freedom. The single primary passes only if the largest of these four p values is below .01 *and* at least 24/32 native AUC values are negative and 24/32 rescues positive. The consistency requirement is about sign, not 24 seeds individually exceeding the practical-effect magnitude.

The four effects are not four alternative routes to success. The conjunction cannot be rescued by choosing the best endpoint, a different reference, another evaluation key, the ratio interval or a secondary control. Nominal 95/99% intervals are paired training-seed intervals. The secondary simultaneous one-sided 95% bounds use a Bonferroni factor of four over these components; this sensitivity is not paper-wide error control. Signs and fixed-seed mean bootstrap results are also preserved in the numerical artifact.

C.3 Reproduce the repair-ratio interval without an RL run

The following is the frozen ratio calculation, made explicit. It reads the already-paired primary rows; the CSV or raw-curve verifier can first confirm them independently.

```
import json
import numpy as np
r = json.load(open(
    "results/summary/p14_single_result-analysis.json"))
Y = np.asarray(r["primary"]["onset_actor_rescue"]["per_seed"])
B = -np.asarray(r["primary"]["native_AUC_harm"]["per_seed"])
rho = Y.mean() / B.mean()
rng = np.random.default_rng(140033)
I = rng.integers(0, 32, size=(20000, 32))
# SAME paired seed indices for both numerator and denominator:
Y_star = Y[I].mean(axis=1)
B_star = B[I].mean(axis=1)
rho_star = Y_star / B_star
ci = np.quantile(rho_star, [0.025, 0.975], method="linear")
assert np.isfinite(rho_star).all()
# No clipping and no replacement by the original mean denominator.
```

The interval is not obtained by dividing separate marginal interval endpoints, independently resampling Y and B, averaging seedwise ratios, or resampling 64 episodes as 32×64 independent trained agents. B and Y share N as well as the original seed history and evaluation tape. Keeping row indices joint preserves that empirical covariance. A toy case Y= cB makes the distinction transparent: a paired resample keeps the ratio exactly c, whereas independent numerator/denominator resamples introduce artificial variation.

The interval is a percentile-bootstrap approximation conditional on the observed design, not an exact coverage theorem for all ratio distributions. Since every observed B is positive, every empirical resampled mean denominator is positive here. A small/uncertain denominator in another setting would require a different discussion; it is not made safe by forcing ratios into 0–100%. The primary statistical decision remains the four-component rule, independent of whether a rounded repair percentage appears large.

D Every original paired primary seed result

Table 11 makes the training-seed unit visible in the PDF. Values are rounded for reading; the linked CSV and raw JSON retain precision. Recomputed trajectories, when matched, duplicate these original seeds and are *not* appended as another 32 rows for inference. One late contrast is positive, even though the mean late effect and all native/rescue consistency requirements pass.

Table 11: All original primary seed effects, in average evaluation-return points. These are the 32 original inferential units, not the later same-seed reconstructions. Signs are retained; no seed or large effect is excluded. Full precision is in the linked CSV/JSON.

seed	native γ_H	late ℓ_H	actor rescue τ_D	reciprocal δ_D
1200	−1.072	−0.316	+0.857	−1.572
1201	−0.983	+0.103	+0.895	−1.653
1202	−2.961	−2.456	+3.059	−2.712
1203	−1.948	−1.859	+1.630	−1.570
1204	−1.426	−1.041	+0.889	−1.436
1205	−1.090	−0.531	+1.215	−1.420
1206	−2.619	−2.216	+2.611	−2.454
1207	−1.359	−0.319	+1.178	−1.508
1208	−2.405	−1.591	+2.258	−2.760
1209	−2.414	−2.128	+1.805	−2.297
1210	−0.991	−0.584	+0.480	−1.136
1211	−2.275	−1.309	+2.584	−1.935
1212	−1.780	−1.266	+1.880	−0.719
1213	−2.071	−0.209	+1.455	−1.803
1214	−1.373	−0.478	+1.159	−1.090
1215	−1.173	−0.872	+1.420	−1.023
1216	−1.101	−0.569	+1.100	−1.959
1217	−2.243	−1.206	+1.944	−2.118
1218	−2.504	−1.831	+1.945	−2.059
1219	−0.585	−0.797	+0.967	−0.935
1220	−1.626	−1.350	+1.565	−1.554
1221	−1.792	−1.403	+1.318	−2.115
1222	−1.864	−1.522	+1.444	−1.182
1223	−2.395	−2.166	+2.117	−1.744
1224	−1.767	−1.681	+1.909	−1.234
1225	−1.465	−0.819	+1.372	−2.060
1226	−1.425	−0.988	+1.088	−0.872
1227	−1.536	−0.463	+1.730	−1.166
1228	−1.781	−1.356	+1.577	−1.279
1229	−2.676	−2.612	+2.294	−1.991
1230	−0.793	−0.409	+1.102	−1.913
1231	−1.609	−1.738	+1.729	−1.100
mean	−1.722	−1.187	+1.580	−1.636

E What can be reanalysed, loaded or recomputed?

Reconstruction status in this working draft. The exact reconstruction is still running; no completed public donor bundle is asserted here. Original mean curves and fingerprint/check records remain available for all 32 primary seeds. The final artifact status will be filled only from verified per-group reports and publication receipts, not from the intended schedule.

Table 12: Reproducibility tiers. "Reconstructed" describes later provenance, not a newly observed independent condition. Static functions are supplied by code, not by executable checkpoint pickles.

material / operation	available information	computation still needed
Original frozen seed-level means	All 11 cells, 21 checkpoints/seed, fork history, both requested evaluation keys, primary/secondary analysis and source hashes.	No RL run to regenerate statistics, consistency, reference alignment or the joint ratio interval.
Original state-integrity record	Recorded all-D/native and all-C/clean full-tree checks; three batched donor numerical fingerprints per group.	A fingerprint cannot reconstruct missing arrays or independently rerun the original memory object.
Later reconstructed donor bundle	P/C^+/D dynamic runner fields with matching original numerical fingerprints; schema, source/environment metadata, serialization audit.	Loading needs the frozen static function/template. Simulating recovery still costs training/evaluation time, but not the prehistory.
Later fixed-batch episode returns	Reconstructed individual returns and call index; means checked against original recorded means.	These are not a new evaluation population or original-retained episode files. A new key requires evaluation computation.
Intermediate/final recovery policy weights	Curves are archived; a policy object at every recovery checkpoint is not separately retained.	Re-run the required continuation from a matching donor to recover those policies or evaluate new tapes.
Whole-process/backend state	Numerical state plus pinned code, not Python function pickles, compiled executables, hardware or an unlimited environment clone.	Rebuild the template/compiler environment; report any mismatch.

E.1 Original and reconstructed data are different provenance classes

The original data-generating run did *not* retain full serialized donor trees. That historical limitation is not rewritten. The later recovery procedure checks P/C^+/D against the original four-seed batch fingerprints, serializes all dynamic numerical leaves, restores them before the unchanged runner proceeds, and compares all five primary/reference trajectories against the original recorded means. The first group additionally repeats all six secondary cells; the other groups retain those original secondary means but do not claim to have rerun them within this reproduction budget. Fresh processes load each checked group into a new template and replay native/restored primary cells with both evaluation keys. Completion counts and public bundle links above are based on checked reports, not assumed from intended execution.

All reconstructed observations use the same 32 seed IDs and protocol. They improve artifact access and auditability, not generalizability or independent statistical sample size. Exact numerical agreement is a reproduction criterion, not a second favourable study to be pooled into the primary. A full four-seed fingerprint is not an individual checkpoint fingerprint, and static apply/optimizer functions are reconstructed from the source rather than recovered from a hash.

E.2 What the integrity audit can and cannot prove

Serialization round-trip equality checks dynamic values against their source donor. The independent load path checks the recorded numerical fingerprint and the source/template rule. Agreement in the declared reconstruction cells checks actual future computation from the deserialized donors; it is not a claim that unrerun secondary cells have also been independently replayed. Fixed-batch episode data additionally permit

exact verification of the 64-episode averaging operation; they do not by themselves establish that uncertainty across alternative evaluation tapes was estimated.

These checks can detect changed donors, stale/corrupt files, a wrong context, skipped fields, misaligned curves and duplicate seed accounting. They cannot prove that a reference is the only scientifically appropriate counterfactual, that information is uniquely encoded in an actor, or that the same intervention works on unseen tasks. A matching deterministic trajectory is also not an external independent implementation of PPO.

F Commands, restoration and failure handling

F.1 Fast statistical reanalysis from the frozen release

Use a working copy and the pinned numerical environment. No JAX training is needed for the following raw-data/statistical checks:

```
python scripts/verify_p14.py
python scripts/build_primary_artifact_index.py
python scripts/verify_p14_uncertainty.py
```

The first script reconstructs all four paired rows, historical lag, shifted t tests, 95/99% intervals, consistency and source/data hashes. The second deterministically exports the 32-row CSV and artifact index. The third reproduces the joint repair-ratio bootstrap without changing the numerical result artifact. The original `scripts/analyze_p14.py` can regenerate the full analysis in a separate working copy and refuses an incomplete cohort.

F.2 Load reconstructed states and replay primary continuations

Each public bundle includes a manifest, P/C/D `.msgpack.gz` files, original and recomputed mean summaries, reconstructed fixed-batch episode returns/call index, pinned requirements and integrity reports. The release index supplies SHA256 values and public download URLs. Verify the bundle and member hashes before deserialization. The format contains numerical state dictionaries, not arbitrary executable pickle objects.

A fresh template is created by the frozen PPO initialization with the recorded environment/configuration and the exact four-seed batch. Flax restoration fills all dynamic fields; static model/optimizer functions come from that template. Verify the restored numerical fingerprint before any continuation. The published replay entry point carries out these operations rather than asking readers to guess how to join parameter/optimizer/RNG files:

```
# Download the seed-group bundle using the indexed URL; verify SHA256.
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
  python scripts/p14_checkpoint_reproduction.py --mode replay \
  --bundle /path/to/s1200to1203.zip \
  --seeds 1200 1201 1202 1203 --cells DDD CDD
```

This regenerates the 21 evaluations of native/restored recovery with both specified keys. It is not a zero-compute replay of logged scores: policies continue learning from the saved initial states. Exact first-group episode-array and frozen-mean comparisons are reported separately from the original primary result.

F.3 Full from-scratch reproduction and clean provenance

The full frozen runner rebuilds prehistory and all 11 cells from seeds. Use a new reconstruction tag or the published reproduction wrapper: an old tag can resume already-complete cells after matching fingerprints, which is not the same as rerunning their trajectories. The wrapper's new tags are excluded from the original primary loader. Do not pool duplicate seed IDs or silently overwrite the original summaries.

Keep the four-seed shapes, source, numerical versions, E=10 grid and fixed evaluation keys. Run sequentially. Each manifest identifies its actual source revision; missing/unrun/partial work is not imputed. The historical experiment's execution deadline is not changed to conduct a later reproduction: the reconstruction has its own explicit operational schedule and no altered scientific hypothesis.

If donor fingerprints, a recorded mean, a checkpoint member hash or a replay array differs, stop and preserve the discrepancy. Do not tune source flags or choose another seed to obtain a match. A software-only CartPole seed 99994 exercises serialization and all 11 cell paths but never enters Asterix inference or generalization counts. The archive records whether each group is complete, matching and independently load-tested; intended future work is not described as accomplished.

F.4 Build the integrated paper and locate the full historical record

The main source includes this appendix and the single donor/reference diagram; it does not require the historical supplement to expose primary estimators or commands. Build from the `paper` directory with Tectonic and three explicit reruns, which avoid its automatic bibliography-rerun inconsistency warning. The companion supplement preserves all old tables/figures, local multiplicity families, interim looks, selected-condition counts, negative/inconclusive primaries and unavailable legacy artifacts. Its earlier PPO/PQN/replay findings must not be relabelled as additional independent support for the one selected Asterix result.